

Some (potentially) helpful formulas:

Student #1

Thermodynamics:

$$dE = dQ + dW$$

$$C_X = \left. \frac{dQ}{dT} \right|_X$$

$$dE = TdS + \sum_i J_i dx_i + \mu dN$$

$$F = E - TS$$

$$G = E - TS - Jx$$

$$\Phi = E - TS - \mu N$$

Kinetic Theory of Gases

$$\frac{d\vec{q}_i}{dt} = \frac{\partial H}{\partial \vec{p}_i}$$

$$\frac{d\vec{p}_i}{dt} = -\frac{\partial H}{\partial \vec{q}_i}$$

$$\{A, B\} = \sum_{\alpha=1}^{3N} \left(\frac{\partial A}{\partial q_\alpha} \cdot \frac{\partial B}{\partial p_\alpha} - \frac{\partial A}{\partial p_\alpha} \cdot \frac{\partial B}{\partial q_\alpha} \right)$$

$$\frac{\partial \rho}{\partial t} = \{H, \rho\}$$

Probability/Mathematical Formulas:

$$\tilde{p}(k) = \langle e^{-ikx} \rangle = \sum_{n=0}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle$$

$$\ln \tilde{p}(k) = \sum_{n=1}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle_c$$

$$\int_0^{\infty} dx x^n e^{-\alpha x} = \frac{n!}{\alpha^{n+1}}$$

$$\int_{-\infty}^{\infty} dx e^{-ikx - \frac{x^2}{2\sigma^2}} = \sqrt{2\pi\sigma^2} e^{-\frac{\sigma^2 k^2}{2}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-ax^2} = \frac{1}{2} \sqrt{\frac{\pi}{a^3}}$$

$$\lim_{x \rightarrow \infty} N! = N \ln N - N$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

Classical Statistical Mechanics

$$p_{(T,x,N)}(\mu_s) = \frac{e^{-\beta H(\mu_s)}}{Z(T,x,N)}$$

$$Z(T,x,N) = \sum_{\mu_s} e^{-\beta H(\mu_s)}$$

$$p_{(T,J,N)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T,J,N)}$$

$$Z(T,J,N) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - Jx)}$$

$$p_{(T,\mu,x)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T,J,N)}$$

$$Q(T,\mu,x) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - \mu N)}$$

$$S = k \ln \Omega$$

$$F = -k_B T \ln Z(T,x,N)$$

$$G = -k_B T \ln Z(T,J,N)$$

$$\Phi = -k_B T \ln Q(T,\mu,x)$$

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$$

$$\langle x \rangle = k_B T \frac{\partial \ln Z}{\partial J}$$

$$\langle N \rangle = \frac{\partial \ln Q}{\partial (\beta \mu)}$$

$$\lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}}$$

Interacting Particles

$$\ln Z = \ln Z_0 + \frac{N(N-1)}{2V} \int d^3\vec{q} [\exp(-\beta V(\vec{q})) - 1] + O\left(\frac{N^3}{V^3}\right)$$

$$\frac{P}{k_B T} = \frac{N}{V} \left[1 + B_2(T) \frac{N}{V} + B_3(T) \frac{N^2}{V^2} + \dots \right]$$

$$f(\vec{q}) = \exp(-\beta V(\vec{q})) - 1$$

$$B_l(T) = -\frac{l-1}{l!} d_l$$

$$(P - an^2)(V - Nb) = Nk_B T$$

$$Z(T, N, V) \approx \frac{(V - N\Omega/2)^2}{N! \lambda^{3N}} \exp \frac{\beta \mu N^2}{2V}$$

Quantum Statistical Mechanics

$$\eta = \begin{cases} +1 & \text{for bosons} \\ -1 & \text{for fermions} \end{cases}$$

$$\ln Q_\eta = -\eta \sum_{\vec{k}} \ln \left[1 - \eta \exp \left(\beta \mu - \beta \epsilon(\vec{k}) \right) \right]^{-1}$$

$$z = e^{\beta \mu}$$

$$N_\eta = \sum_{\vec{k}} \frac{1}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$E_\eta = \sum_{\vec{k}} \frac{\epsilon(\vec{k})}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$\beta P_\eta = \frac{g}{\lambda^3} f_{5/2}^\eta(z)$$

$$n_\eta = \frac{g}{\lambda^3} f_{3/2}^\eta(z)$$

$$\epsilon_\eta = \frac{3}{2} P_\eta$$

$$f_m^\eta(z) = \frac{1}{(m-1)!} \int_0^\infty \frac{dx}{x^{m-1}} z^{-1} e^x - \eta = \sum_{\alpha=1}^\infty \eta^{\alpha+1} \frac{z^\alpha}{\alpha^m}$$

$$\frac{d}{dz} f_m^\eta(z) = \frac{1}{z} f_{m-1}^\eta(z)$$

Problem 1 (35 points)

Consider a polymer formed by connecting N disc-shaped molecules into a one-dimensional chain. Each molecule can align along either its long axis (of length $2a$) or short axis (of length a). Furthermore, when laying on its long axis there are two configuration it may assume, whereas when lying along its short axis there is only one possible configuration. The energy of the monomer aligned along its shorter axis is higher by ϵ , that is, the total energy of the system is $H = \epsilon U$ where U is the number of monomers standing up.

a (10 points) Calculate the partition function $Z(T, N)$ of the polymer.

1 Soln
Using $Z(T, J, N) = \sum_{H_s} e^{-\beta(H_s - Jx)}$

For the length of $2a$, it can either be down or up

for $2a = N^2$, possible ways

For the a i.e short axis it can only up or down

$$\frac{N(N-1)}{2} \text{ possible ways.}$$

b (15 points) Calculate the average length $\langle L(T, N) \rangle$ and the variance of the length $\langle L(T, N)^2 \rangle_c$ of the polymer.

2 Soln

Where

$$L = x$$

$$\langle L \rangle = k_B T \frac{\partial \ln Z}{\partial J}$$

$$\text{Where, } Z(T, J, N) = \sum_{H_s} e^{-\beta(H_s - Jx)}$$

Where $H = \epsilon U$

$$Z(T, J, N) = \sum_U e^{-\beta(\epsilon U - Jx)}$$

Take the natural log

$$\begin{aligned} \ln Z &= \ln -\beta(\epsilon U - Jx) \\ &= -\ln \beta \epsilon U + \ln \beta Jx \end{aligned}$$

$$\frac{\ln Z}{\partial J} = \frac{\beta}{Jx}$$

$$\langle L \rangle = k_B T \left(\frac{\beta}{Jx} \right)$$

$$= k_B T \left(\frac{1}{k_B T} \right) = \frac{1}{Jx}$$

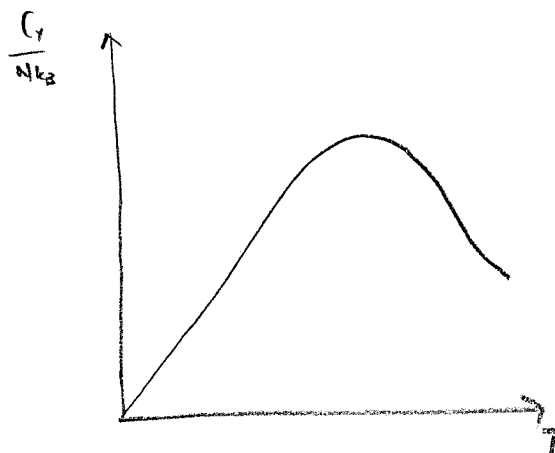
c (10 points) Determine the energy per monomer and heat capacity per monomer. Sketch the heat capacity as a function of temperature.

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$$(i) E = - \frac{\partial \ln Z}{\partial \beta}$$

(ii)

$$C = \frac{dE}{dT}$$



Problem 2 (35 points)

The pressure P of a gas is related to its density $n = N/V$ and temperature T by the truncated expansion:

$$P = k_B T n - \frac{b}{2} n^2 + \frac{c}{6} n^3$$

where b and c are assumed to be positive temperature-independent constants.

- a (15 points) Locate the critical temperature T_c below which the equation must be invalid, and the corresponding density n_c and pressure P_c of the critical point. Find the ratio $k_B T_c n_c / P_c$

Soln

Given

$$P = k_B T n - \frac{b}{2} n^2 + \frac{c}{6} n^3 \quad (1)$$

where at critical temperature

$$\frac{dP}{dn} = 0$$

$$\frac{d^2 P}{dn^2} = 0$$

lets find the derivative of eqn (1) w.r.t. n

$$\frac{dP}{dn} = k_B T - b n + \frac{c n^2}{2}$$

$$\text{when } \frac{dP}{dn} = 0$$

$$k_B T - b n + \frac{c n^2}{2} = 0$$

$$\frac{d^2 P}{dn^2} = -b + c n, \text{ when } \frac{d^2 P}{dn^2} = 0$$

$$0 = -b + c n$$

$$n_c = \frac{b}{c} \quad \checkmark$$

$$k_B T_c - \frac{b^2}{c} + \frac{c b^2}{2 c^2} \quad \text{math - 1}$$

$$k_B T_c - \frac{b^2}{c} + \frac{b^2}{2c} = k_B T_c = \frac{b^2}{2c}$$

$$T_c = \frac{b^2}{2 k_B c}$$

lets find P_c

$$P_c = k_B T_c n_c - \frac{b}{2} n_c^2 + \frac{c}{6} n_c^3$$

$$\begin{aligned} P_c &= k_B \frac{b^2}{2 k_B c} \times \frac{b}{c} - \frac{b}{2} \times \frac{b^2}{c^2} + \frac{c}{6} \frac{b^3}{c^3} \\ &= \frac{b^2}{2 c^2} \times \frac{b^3}{2 c^2} + \frac{b^3}{6 c^2} \end{aligned}$$

$$\frac{k_B T_c n_c}{P_c} = \frac{\frac{k_B b^2}{2 c k} \frac{b}{c}}{\frac{b^3}{6 c^2}} = \frac{1}{2} \times \frac{6 c^2}{b^2} = 3$$

b (10 points) Calculate the isothermal compressibility $\kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \Big|_T$ and sketch its behavior as a function of T for $n = n_c$.

Soln

using $V = \frac{N}{n}$

$$\frac{dV}{dn} = -\frac{N}{n^2}$$

$$dV = -\frac{V}{n} dn$$

$$\frac{\partial V}{\partial P} = -\frac{N}{n^2} \frac{\partial n}{\partial P}$$

$$\kappa_T = -\frac{1}{V} \left(-\frac{N}{n^2} \frac{\partial n}{\partial P} \right)$$

$$= \frac{1}{n} \left(\frac{\partial n}{\partial P} \right) \Big|_T$$

Using equation of state

$$P = kTn - \frac{b}{2}n^2 + \frac{c}{2}n^3$$

$$\kappa_T = \frac{1}{n} \left(\frac{1}{kT - bn + \frac{3c}{2}n^2} \right) = \frac{1}{n(kT - bn + \frac{3c}{2}n^2)}$$

c (10 points) On the critical isotherm give an expression for $(P - P_c)$ as a function of $(n - n_c)$

Soln

Given $P = kTn - \frac{b}{2}n^2 + \frac{c}{2}n^3$

$$\frac{dP}{dn} = kT - bn + \frac{3c}{2}n^2$$

$$kT - bn + \frac{3c}{2}n^2 = 0$$

Since $\frac{dP}{dn} = 0$

$$\frac{d^2P}{dn^2} = -b + 3cn$$

When $\frac{d^2P}{dn^2} = 0$

$$n_c = \frac{b}{3c}$$

Solving for T_c

$$kT_c - bn + \frac{3c}{2}n^2 = 0$$

$$n_c = \frac{b}{3c}, kT_c - \frac{b^2}{3c} + \frac{3c}{2} \left(\frac{b}{3c} \right)^2 = 0$$

$$kT_c - \frac{b^2}{3c} + \frac{b^2}{2c} = 0$$

where,

$$T_c = \frac{b^2}{2kc}$$

Let's find P_c

$$P_c = \frac{b^3}{6c^2} \text{ from the previous question}$$

Let's find $P - P_c$ by plugging in P and P_c

$$P - P_c = \frac{c}{b} (n - n_c)^3$$

$$P - P_c = \left(kTn - \frac{b}{2}n^2 + \frac{c}{2}n^3 \right) - \left(kT_c n_c - \frac{b}{2}n_c^2 + \frac{c}{2}n_c^3 \right)$$

Problem 3 (30 points + 10 bonus)

Molecular hydrogen exists in two states: ortho and para.

a (20 points) In para-hydrogen, the two electrons of H_2 form a singlet (antisymmetric) state. Therefore, the orbital angular momentum can have only even values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1)$$

for $l = 0, 2, 4, \dots$

In contrast, in ortho-hydrogen, the electrons are in a triply degenerate symmetric state. Therefore, the orbital angular momentum can have only odd values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1)$$

for $l = 1, 3, 5, \dots$

Compute the rotational partition function for both para-hydrogen and ortho-hydrogen, and evaluate both of their low and high temperature limits.

$$Z_p = \sum_{l=0,2,4,\dots} (2l+1) e^{-\beta \frac{\hbar^2}{2I} l(l+1)}$$

When $l = 2n$, with $n = 0, 1, 2$

$$Z_p = \sum_{n=0}^{\infty} (4n+1) e^{-\beta \frac{\hbar^2}{2I} 2n(2n+1)}$$

In low temperature, the limit $\beta \rightarrow \infty$, then $e^{-\beta \frac{\hbar^2}{2I} l(l+1)}$ becomes very large for higher values of l

For $n=0$ and corresponding $l=0$

$$E_n = \frac{\hbar^2}{2I} l(l+1), E_0 = \frac{\hbar^2}{2I} 0(0+1) = 0 \text{ and } 2(2n)+1 = 4n+1$$

$$0+1=1$$

Then the first term in Z_p

$$Z_p = 1 e^{-\beta \cdot 0} = 1$$

For $n=1, l=2$

$$E_1 = \frac{\hbar^2}{2I} 2(2+1) = 3 \frac{\hbar^2}{I}, 2(2n)+1 = 4+1=5$$

$$Z_p = 5 e^{-3\beta \frac{\hbar^2}{I}}$$

For $n=2, l=4$

$$E_2 = \frac{\hbar^2}{2I} 4(4+1) = 10 \frac{\hbar^2}{I}, 2(4)+1=9$$

$$Z_p = 9 e^{-10\beta \frac{\hbar^2}{I}}$$

$$Z_p \approx 1 + 5 e^{-3\beta \frac{\hbar^2}{I}} + 9 e^{-10\beta \frac{\hbar^2}{I}}$$

For temperature $\beta \rightarrow 0$ and the spacing between energy is small compared to $k_B T$ and many rotational states are populated

$$Z_p = \sum_{n=0}^{\infty} (4n+1) e^{-\beta \frac{\hbar^2}{2I} 2n(2n+1)}$$

$$Z_p \approx \int_0^{\infty} (4x+1) e^{-\frac{\beta \hbar^2}{2I} 2x(2x+1)} dx$$

$$Z_p \approx \int_0^{\infty} (4L+1) e^{-\beta \frac{\hbar^2}{2I} (2L)(2L+1)} dL$$

$$Z_p \approx \frac{1}{\beta \hbar^2} = \frac{k_B T}{\hbar^2}$$

(ii)

$$H = \frac{\hbar^2}{2I} L(L+1)$$

$$Z_o = \sum_{L=1,3,5,\dots} (2L+1) e^{-\beta \frac{\hbar^2}{2I} L(L+1)}$$

Since the electrons are in a triply degenerate symmetric state

$$Z_o = 3 \sum_{L=1,3,5,\dots} (2L+1) e^{-\beta \frac{\hbar^2}{2I} L(L+1)}$$

$L = 2n+1$, since $L = \text{odd}$

$$Z_o = 3 \sum_{n=0}^{\infty} 2(2n+1) e^{-\beta \frac{\hbar^2}{2I} (2n+1)(2n+2)}$$

$$= 3 \sum_{n=0}^{\infty} (4n+2) e^{-\beta \frac{\hbar^2}{2I} (2n+1)(2n+2)}$$

For low temperature ($\beta \rightarrow \infty$) and only the first terms in the sum contribute significantly to Z_o

The lowest energy level corresponds to $L=1, n=0$

$$Z_0 \approx 3(3)e^{-\beta \hbar^2/2I} \approx 9e^{-\beta \hbar^2/2I}$$

For high temperatures $\beta \rightarrow 0$

$$Z_0 \approx \int_0^\infty 3(2L+1) e^{-\beta \hbar^2/2I} L(L+1) dL$$

$$\approx \frac{3I}{\beta \hbar^2}$$

- b (10 points) For an equilibrium gas of N hydrogen molecules composed of N_p para and $N_o = N - N_p$ ortho molecules, calculate the partition function of the complete system. (you may ignore vibrational degrees of freedom)

Soln

$$Z = \sum_{N_p=0}^N \frac{1}{N_p!(N-N_p)!} Z_p^{N_p} Z_o^{N-N_p}$$

Using binomial expansion

$$(Z_p + Z_o)^N = \sum_{N_p=0}^N \frac{1}{N_p!(N-N_p)!} Z_p^{N_p} Z_o^{N-N_p}$$

Then

$$Z = (Z_p + Z_o)^N \times \frac{1}{N!}$$

- c (10 points) Compute an expression for the rotational contribution to the internal energy $\langle E_{rot} \rangle$. Determine its behavior at the high and low temperature limits. Note that due to the large transition times between ortho- and para-hydrogen, in most circumstances the mixture is not actually in equilibrium.

Soln

$$\langle E_{rot} \rangle = - \frac{\partial \ln Z}{\partial \beta}$$

where $Z = (Z_p + Z_o)^N \times \frac{1}{N!}$

$$\langle E_{rot} \rangle = -N \frac{\partial \ln(Z_o + Z_p)}{\partial \beta}$$

at low temperature

$$Z_o + Z_p \approx 1 + 9e^{-\beta \hbar^2 / I}$$

taking the derivative wrt. β

$$\langle E_{rot} \rangle = \frac{9N\hbar^2}{I} e^{-\beta \hbar^2 / I}$$

at high temp ($\beta \rightarrow 0$)

$$Z_o + Z_p \approx \frac{4I}{\beta \hbar^2}$$

Therefore $\langle E_{rot} \rangle \approx N k_B T$